



Learning Python - Sheet #8

You write the game!

This time, you're going to write your own game. This sheet is going to guide you through it, but you will be doing all the hard work!

1. Objective

We want the computer to pick a combination of 5 digits. The player can enter a combination. The computer will tell for each digit whether it's part of the combination and if so, whether it's at the right place or not. The player has 5 tries.

For example:

```
> 12345
. - X-
```

Would mean that the player entered "12345" and that the computer answered that:

- 2 and 5 are part of the combination, but not at the right place
- 4 is part of the combination and at the right place

(For example, the combination could be 25849).

2. Steps

Create a new file (**File > New File**)

2.a - Create the combination

The function should return a string composed of the 5 digits of the combination.

There are multiple options:

- you can pick a number between 0 and 99999. If it's less than 10000, we still want 5 digits. The easiest way is to make it a string with size 5 and leading zeroes. This can be achieved with **"%05d" % number**
- you can create a **DIGITS** constant with all allowed numbers (0 to 9) and call **random.choice(DIGITS)** 5 times to pick the 5 numbers. Each selected digit can then be added to a string.

2.b - Ask for a combination

The function should ask for a combination, verify that it's 5 digits (you can check that it's a string of length 5 and that it can be converted to an integer without error, or check that each char is a digit).

2.c - Build the answer

The function should take 2 parameters: the real combination and the player's tentative. It should go through the player's tentative and, for each digit, compare it to the corresponding digit in the combination. If it's the same, then add a X. Otherwise, check all the digits in the combination. If one matches, add a -. Otherwise, add a space.

2.d - Glue it all together

With all these functions, you can code the game:

- create the secret combination
- then repeatedly:
 - ask for the player's combination
 - build the answer
 - display the answer
 - if both combinations are the same, the player won
 - if it's the last try, the player lost



Practice to progress:

You can play a bit with the game, and see what can be improved:

- can you make it start a new game as soon as you find the number? **Tip:** to make it easier you may want to create a function for a round of the game
- can you make the game have multiple levels with different difficulties? (allow digits and letters in the combination, have a longer combination, have more or less tries, ...)

As you play, do you find ideas on how to make this game better? That's the great thing about programming: you can always come up with new ideas to make your program better.

